Milestone 1 | Dataset memo

IE 7615 - Group 1 - Project 2

Dataset and preparation

The dataset began with 1,000 images from COCO 2017, each paired with one caption. The selection covered all 80 object categories and retained images labeled Attribution or Attribution-ShareAlike. A fixed random seed (42) made the selection reproducible.

Captions were cleaned to remove spacing problems and repeated text, with a 5-35-word filter. Images were checked for readability and minimum size, converted to RGB, and resized to 512 x 512 with padding to preserve the full scene. Normalization to [-1, 1] was verified after loading; CLIP used its own processor.

Review and improvements

All 1,000 final image-caption pairs underwent an iterative verification loop. Unclear cases were examined at full resolution, captions were corrected, and flagged pairs were checked again. Review decisions were recorded so each change could be traced.

This process revised 133 captions. Most changes selected a clearer existing COCO caption, while 20 corrected factual details.

Two images were replaced. One showed a close-up facial injury. Its graphic content fell outside the intended everyday scenes, and its caption made claims about the injury and the person that the image could not establish. The other showed almost the same cat, setting and camera angle as another selected image. Replacing it reduced repetition. Both replacement pairs were reviewed.

Automated checks confirmed that the final images loaded correctly, had the required dimensions and contained no exact duplicates. Similarity checks also flagged possible near duplicates for visual inspection. Repeating the preparation reproduced the dataset manifest.

Measure	Result
Final image-caption pairs	1,000
Visually reviewed final pairs	1,000
Caption revisions / factual edits	133 / 20
Images replaced / exact duplicates	2 / 0
Development / validation / evaluation	800 / 100 / 100
Attribution / Attribution-ShareAlike	668 / 332
Observed caption length	7-24 words

Ethical considerations and limitations

The review favored descriptions of visible details and removed unsupported assumptions about people, actions and circumstances. Original captions, corrections, image sources and license information were retained so changes could be traced. COCO annotations were licensed under CC BY 4.0. Downloaded images were excluded from Git.

People appeared frequently, so the subset did not represent every subject equally. The review could still miss caption errors or subtle near duplicates, and padding added borders. Separate dataset splits also could not rule out overlap with pretrained model data. These checks improved consistency without establishing that every caption was correct.

COCO source and terms. Sources and license links were recorded per image in data/manifest.csv.